CS23532 - COMPUTER NETWORKS LABORATORY

EXPERIMENT 10

AIM:

To design and configure an internetwork using routers in CISCO PACKET TRACER simulator.

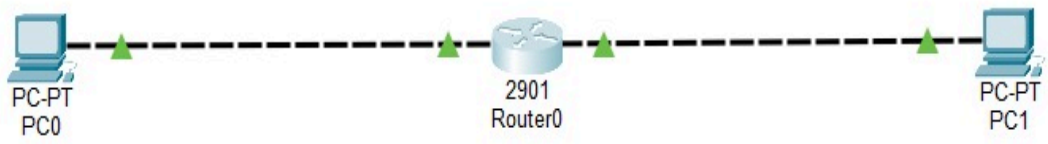
DESCRIPTION:

- **Internetworking:** Connects two or more distinct local area networks (LANs) operating on different network subnets via a central router.
- **Router Functionality:** Routers inspect incoming packet IP headers and use routing tables to forward packets across different network interfaces.
- **Packet Tracing:** Using Packet Tracer's Simulation mode allows step-by-step observation of ICMP Echo Requests (PDUs) traversing from a source host to the router gateway and reaching the destination host across network boundaries.

STEP-BY-STEP PROCEDURE

Task 1: Building the Internetwork Topology

1. Open Cisco Packet Tracer and create a new blank workspace.
2. Add components: **1 Router (2901)** and **2 Generic PCs (PC0 and PC1)**.
3. Connect devices using **Copper Crossover Cables** (dashed black line ---):
    - PC0 (FastEthernet0) → Router0 (GigabitEthernet0/0)
    - PC1 (FastEthernet0) → Router0 (GigabitEthernet0/1)



Task 2: Router Gateway Configuration

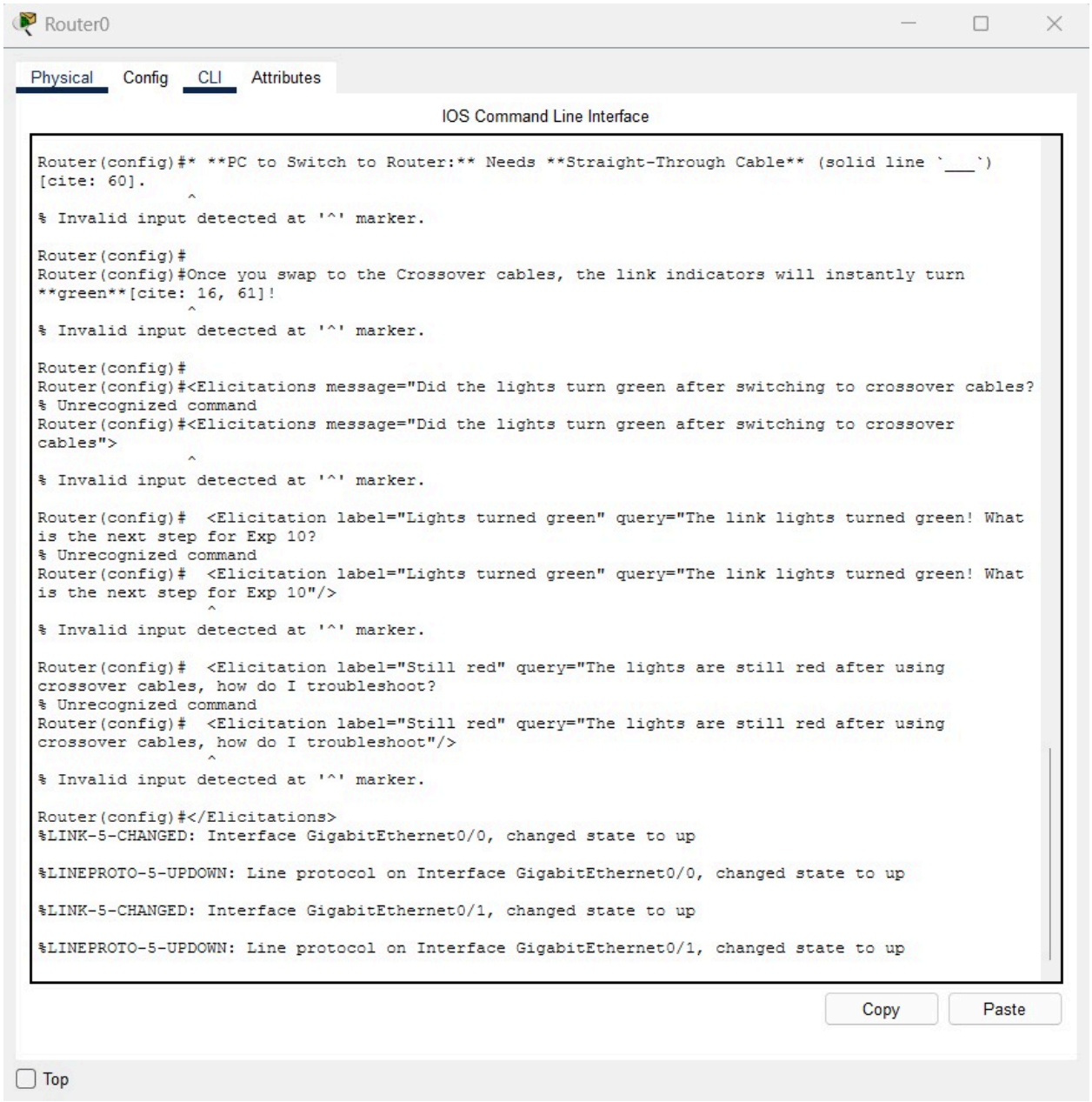
1. Click **Router0**, open the **CLI** tab, type **no** if asked for initial configuration, and hit Enter.
2. Enter the interface IP configuration commands:
3. Plaintext

```
Shell
enable
configure terminal

interface GigabitEthernet0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface GigabitEthernet0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

Observe link status indicators changing to **green**.



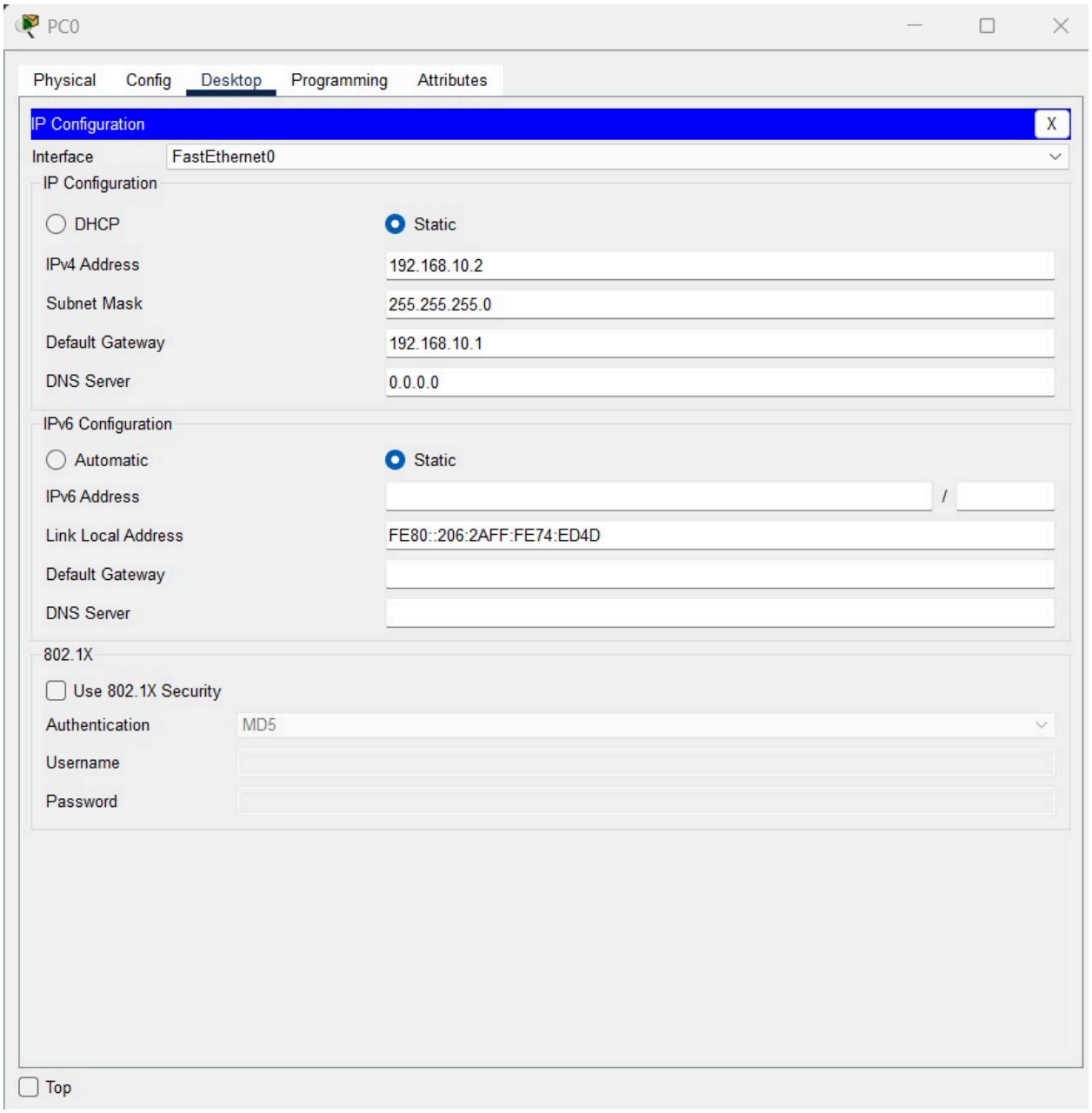
Task 3: Host IP & Gateway Configuration

Click each PC, open **Desktop** → **IP Configuration**, select **Static**, and set the parameters:

- **PC0 Settings (Subnet A: 192.168.10.0/24):**
    - IP Address: 192.168.10.2
    - Subnet Mask: 255.255.255.0

- Default Gateway: 192.168.10.1

- **PC1 Settings (Subnet B: 192.168.20.0/24):**
  - IP Address: 192.168.20.2
  - Subnet Mask: 255.255.255.0
  - Default Gateway: 192.168.20.1

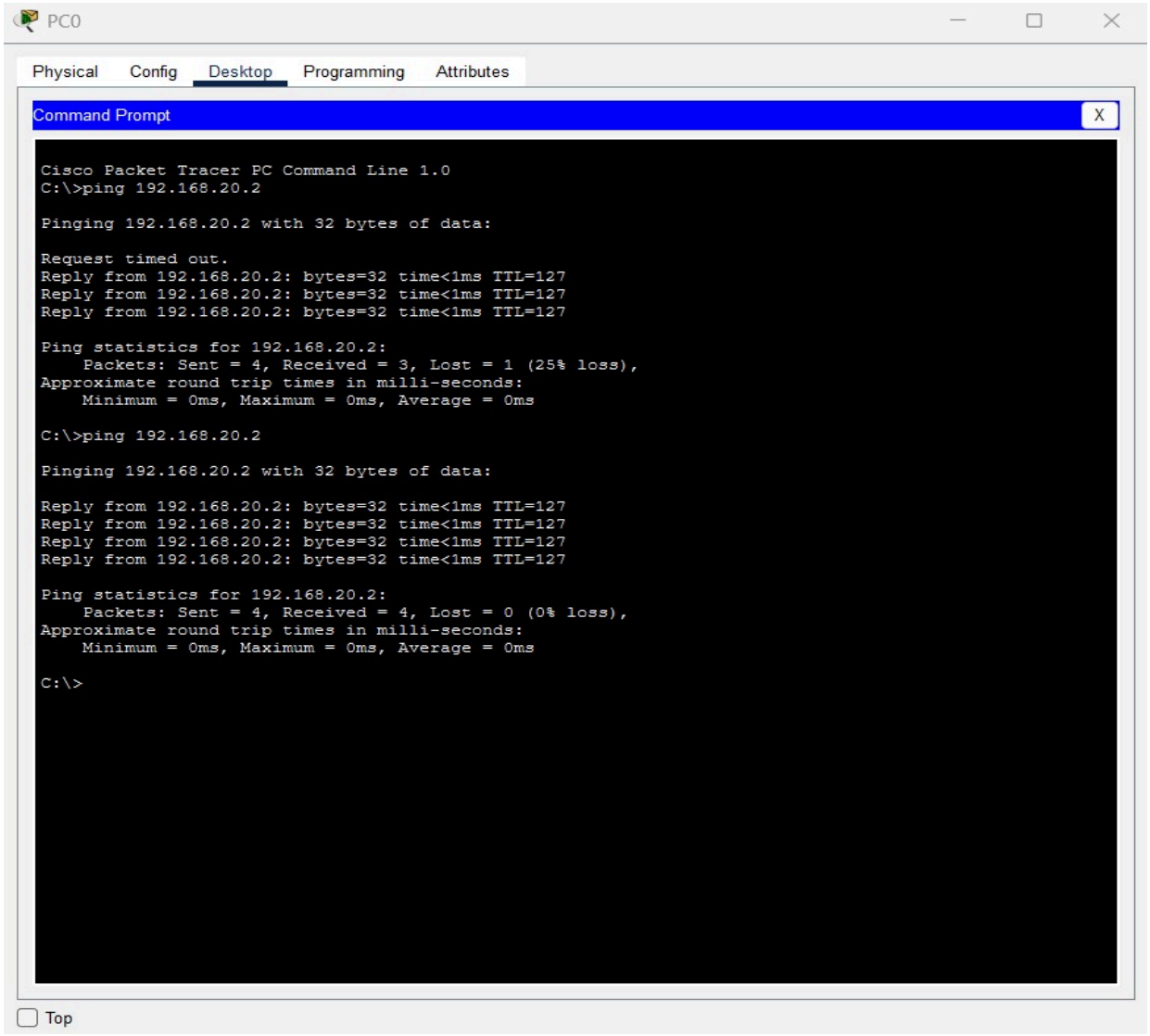


Task 4: Connectivity Testing & Verification

1. Open **Command Prompt** on PC0 (192.168.10.2).
2. Run cross-subnet ping:
3. Plaintext

```
Shell
ping 192.168.20.2
```

4. Verify successful replies with 0% packet loss.
5. Switch to **Simulation Mode**, add a **Simple PDU** between PC0 and PC1, and click **Play / Capture Forward** to trace ICMP traffic across Router0.



ADDRESSING TABLE

Device Name	Interface	IP Address	Subnet Mask	Default Gateway	Connected Subnet
Router 0	GigabitEthernet0/0	192.168.10.1	255.255.255.0	*N/A (Gateway)*	192.168.10.0/24
Router 0	GigabitEthernet0/1	192.168.20.1	255.255.255.0	*N/A (Gateway)*	192.168.20.0/24
PC0	FastEthernet0	192.168.10.2	255.255.255.0	192.168.10.1	192.168.10.0/24
PC1	FastEthernet0	192.168.20.2	255.255.255.0	192.168.20.1	192.168.20.0/24

Result: The experiment to design and configure a simple internetwork using a router in Cisco Packet Tracer simulator has been constructed, configured, and tested successfully.